

PROVISIONAL PATENT APPLICATION

Title

Spatial Logic Computation Systems, Circuits, and Hardware Architectures

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Field of the Invention

The present invention relates generally to computation architectures, logic circuits, and information processing systems, and more particularly to **spatial logic computation, hardware embodiments of spatial state spaces, and networked spatial computation systems** applicable to artificial intelligence, biological modeling, physics simulation, optimization, and education.

Background of the Invention

Modern computing systems are fundamentally based on linear, sequential logic architectures derived from Boolean algebra and the von Neumann model. In such systems, computation is performed through ordered instruction execution over linear memory representations. While these architectures are effective for arithmetic and symbolic manipulation, they exhibit substantial inefficiencies and conceptual limitations when applied to inherently spatial, relational, or high-dimensional domains.

In practice, spatial phenomena—including physical systems, biological systems, learning systems, and complex networks—are simulated on linear machines by encoding spatial relationships into serialized data structures and time-stepped algorithms. This approach imposes significant representational overhead, computational inefficiency, and conceptual distortion, requiring space to be continuously flattened, discretized, and reassembled in software.

Current approaches to artificial intelligence, biological simulation, physics engines, and optimization systems therefore rely on layered abstractions that emulate spatial structure rather than embodying it. These approaches require increasing computational resources

while failing to natively capture the underlying geometry, symmetry, and constraint-based structure of the systems being modeled.

Accordingly, there exists a need for a new class of computation architecture in which **spatial structure itself is the computational substrate**, rather than an emergent property reconstructed through software.

Summary of the Invention

The present invention provides **spatial logic computation systems** in which computation is performed through **state transitions within a predefined spatial topology**, implemented directly in logic circuits and hardware architectures.

In contrast to linear instruction-based computation, the disclosed systems treat information as **formation within a constrained space of possibilities**, where computation consists of transformations, transitions, and stabilizations of spatial states.

In one aspect, the invention comprises a **spatial logic circuit** whose permissible states and transitions are defined by a fixed geometric or topological structure. Computation proceeds by moving between allowable states within this structure according to deterministic or probabilistic rules encoded in hardware.

In another aspect, the invention comprises **networked spatial logic units**, wherein multiple spatial logic circuits are coupled such that their state transitions influence one another, forming higher-order computational fields capable of learning, optimization, and adaptive behavior.

In further aspects, the invention includes software emulators, programmable hardware implementations (e.g., FPGA), and fixed silicon embodiments (ASIC), as well as compilers, interfaces, and educational systems for interacting with spatial logic computation.

Core Conceptual Framework

Information as Formation

In the disclosed invention, information is defined not as a discrete symbolic value, but as a **configuration within a space of possible formations**. Two fundamental sets are recognized:

1. The set of **all possible formations** permitted by a spatial topology.

2. The set of **optimal, stable, or selected formations** within that space.

Computation consists of transitions within and between these sets, governed by structural constraints rather than linear instruction sequences.

Spatial Logic Architecture

Spatial State Space

A spatial logic system according to the invention defines:

- A finite or infinite **state space**
- A geometric or topological structure constraining allowable states
- A set of **legal transitions** between states

Examples of spatial structures include, but are not limited to:

- Polyhedral state spaces
- Graph-based lattices
- Symmetry-constrained networks
- Topological manifolds
- Higher-dimensional relational spaces

Logic as Geometry

Unlike Boolean logic, which evaluates truth values through linear operations, spatial logic evaluates system evolution through **geometric constraints**, adjacency relations, and symmetry-preserving transformations.

Logic operations may include:

- State rotations
- Reflections
- Transpositions
- Permutations
- Constraint-preserving moves
- Energy- or cost-minimizing transitions

Hardware Embodiments

Spatial Logic Circuits

In one embodiment, spatial logic is implemented directly in hardware as a circuit whose nodes represent allowable states and whose connections represent allowable transitions. Circuit behavior enforces spatial constraints natively, without requiring serialized instruction execution.

Such circuits may include:

- Custom logic gates encoding spatial transition rules
- Fixed wiring topologies reflecting spatial adjacency
- Constraint-enforcing elements limiting illegal transitions

Programmable Implementations

In some embodiments, spatial logic systems are implemented on:

- Field-programmable gate arrays (FPGAs)
- Reconfigurable logic fabrics
- Hybrid CPU/spatial coprocessor architectures

These embodiments enable experimentation, simulation, and development prior to fixed silicon fabrication.

Fixed Silicon (ASIC)

In preferred embodiments, spatial logic architectures are implemented as application-specific integrated circuits (ASICs), enabling:

- High efficiency
- Low power consumption
- Native parallelism
- Real-time spatial computation

Networked Spatial Computation

Multiple spatial logic units may be interconnected to form **spatial computation networks**, wherein:

- Each unit maintains its own constrained state space
- Coupling rules define inter-unit influence
- Global behavior emerges from local spatial interactions

Such networks may exhibit:

- Learning behavior
- Adaptive optimization
- Self-organization
- Error tolerance
- Robustness to noise

Software and Emulation

While hardware embodiments are preferred, the invention also includes:

- Software emulators of spatial logic circuits
- Compilers translating spatial problems into spatial logic representations
- Interfaces between linear computing systems and spatial logic hardware

Software implementations serve as:

- Development tools
- Educational platforms
- Compatibility layers with existing systems

Applications

The disclosed invention is applicable to a wide range of domains, including but not limited to:

- Artificial intelligence and machine learning
- Biological modeling and molecular simulation

- Physics simulation and computational geometry
 - Optimization and constraint satisfaction
 - Cryptography and security
 - Education and human learning systems
 - Scientific visualization
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Advantages Over Prior Art

The disclosed systems offer several advantages over existing computing architectures:

- Native spatial computation without linear emulation
 - Reduced representational overhead
 - Improved efficiency for relational problems
 - Hardware-level embodiment of symmetry and constraint
 - Improved alignment with physical and biological systems
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Conclusion

The present invention introduces a fundamentally new class of computation in which **space itself is the logic substrate**. By embedding spatial structure directly into hardware and circuits, the invention enables forms of computation that are inefficient or infeasible on linear architectures, opening new pathways for intelligence, modeling, and learning systems.